Machine Learning for AIOT

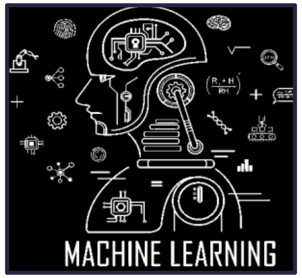
ML Process → Data + Algorithm + Training = ML Model

Historical Data
- Raw Data

ML Algorithm
- Learns hidden relationships

Trained Model
- Prediction engine

New Live Data
- Real-time prediction



Machine Learning enables systems to:
- Learn from historical data
- Identify hidden patterns
- Make predictions
- Improve decision-making automatically

Traditional IoT Systems
- Monitor devices
- Generate fixed alerts
- Depend on hardcoded rules

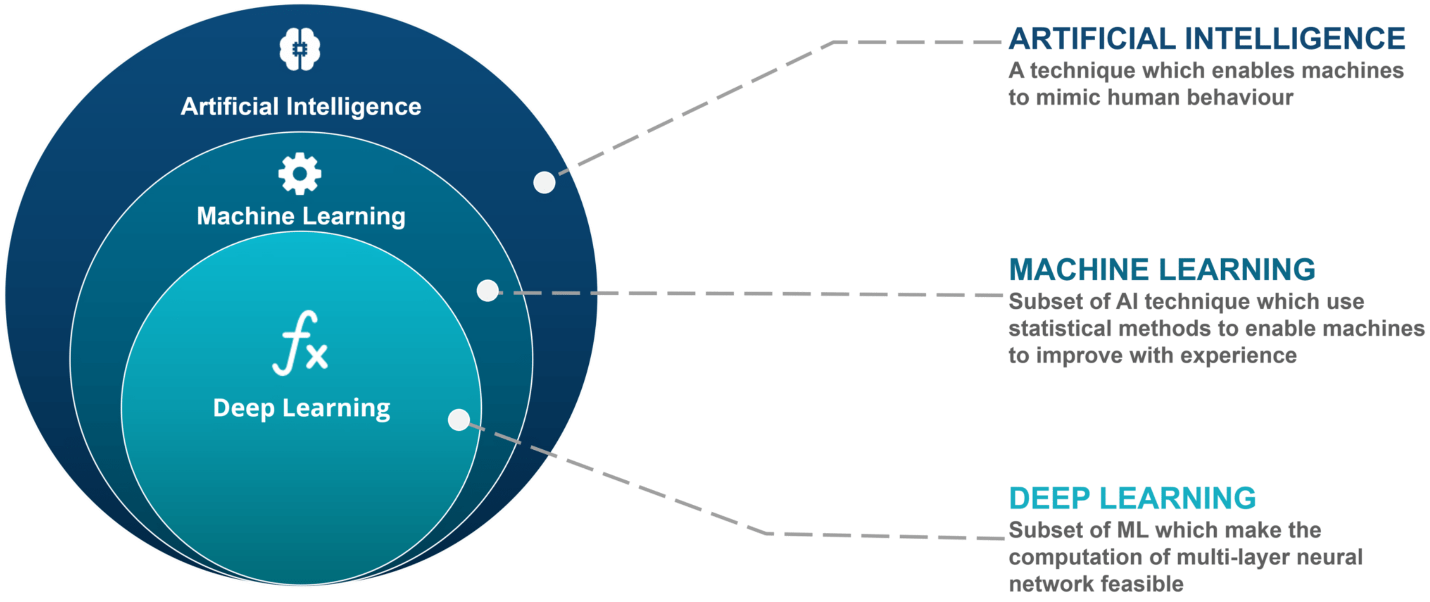
AIoT Systems
- Predict failures
- Detect anomalies
- Learn operational behaviour
- Adapt dynamically
- autonomous decision-making

AIoT Perspective

IoT Devices Generate:
- Temperature data
- Vibration data
- Pressure data
- Sensor telemetry

ML Converts this Data Into:
- Predictions
- Anomaly detection
- Intelligent alerts
- Automation

Machine Learning Vs Deep Learning



- All deep learning is machine learning, not all machine learning is deep learning
- Fundamental difference lies in **how they learn**: traditional machine learning requires humans to manually extract data features, while deep learning uses multi-layered neural networks to automatically discover features from raw data

- **Machine Learning Approach:** A human defines rules for what a car looks like (e.g., wheels are round, windshields are reflective). The algorithm uses these specific inputs to identify a car.
- **Deep Learning Approach:** A neural network looks at 100,000 raw images of cars. It figures out on its own what constitutes a car by analyzing pixel patterns.

Machine Learning Vs Deep Learning

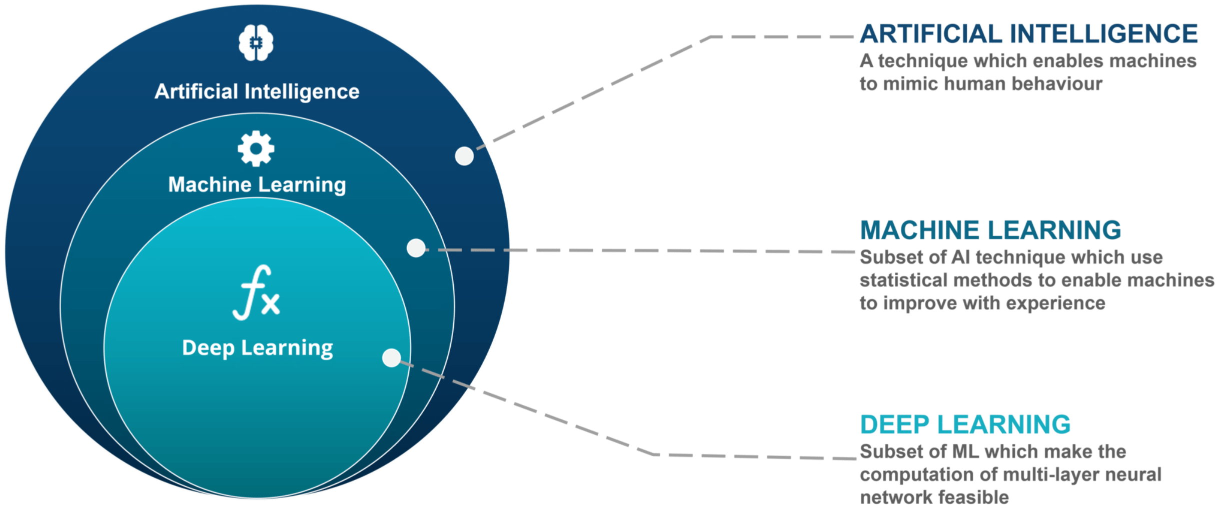
In AIoT systems, both Machine Learning and Deep Learning have important roles.

MACHINE LEARNING

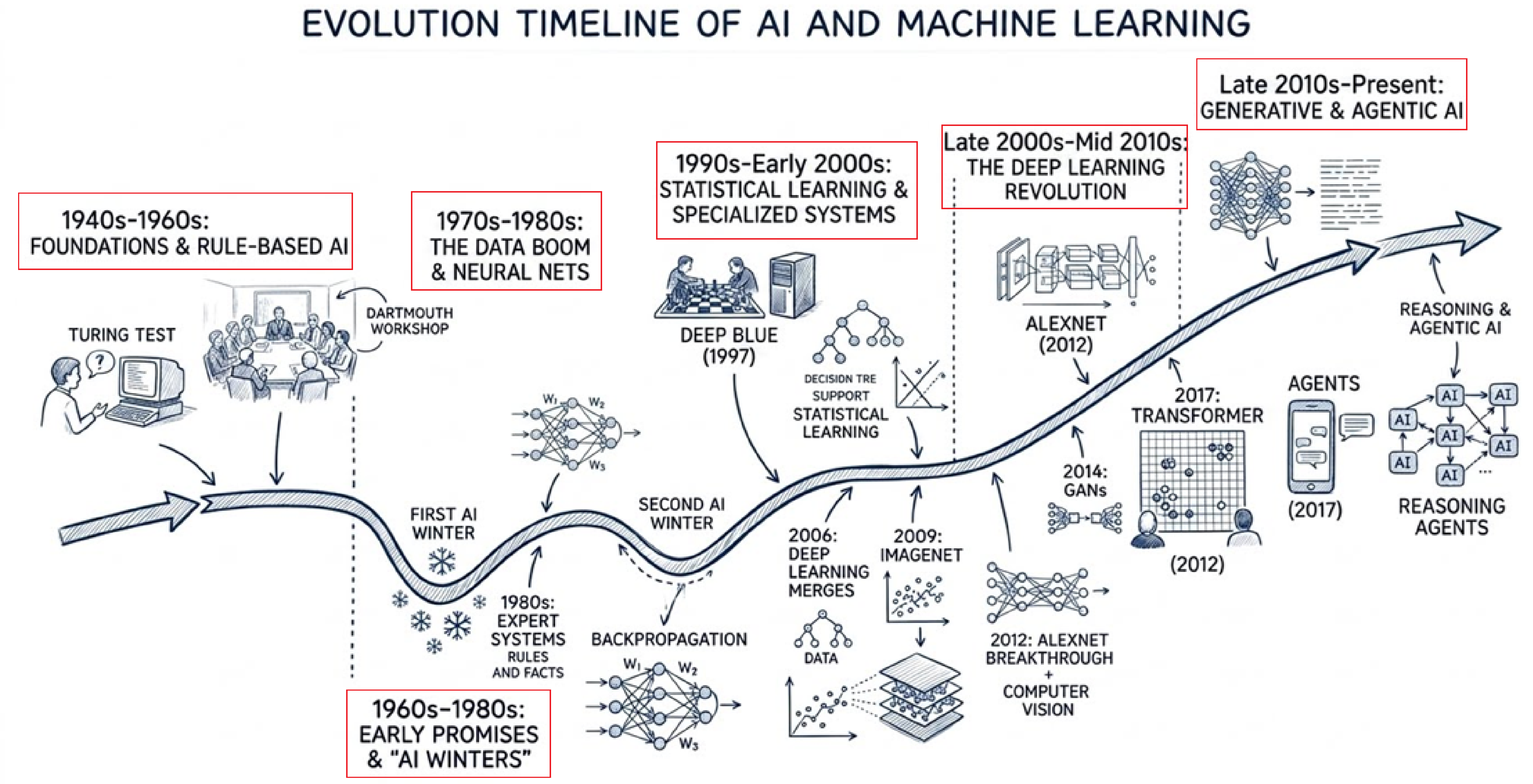
Traditional Machine Learning is commonly used for structured sensor data such as temperature, pressure, vibration, energy usage, and predictive maintenance applications.

DEEP LEARNING

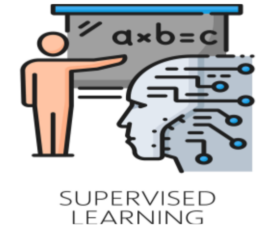
Deep Learning becomes highly useful when AIoT systems involve computer vision, video analytics, speech processing, autonomous systems, smart surveillance, or advanced pattern recognition



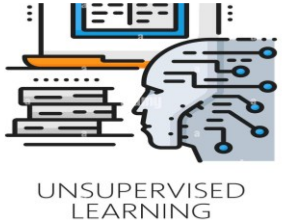
AI-ML Evolution



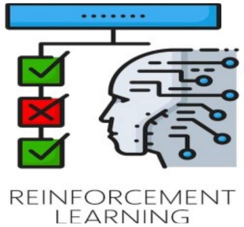
Types of Machine Learning in IoT



Supervised Learning : Learning from labeled historical data
Key Idea : Model learns using input. Used for prediction and classification
Example : Humidity + Pressure → Predict Temperature
AIoT Use Case : Predictive maintenance, Temperature predictions, Energy forecasting, Fault prediction, Quality inspection



Unsupervised Learning : Finding hidden patterns automatically
Key Idea :- No predefined answers. Model discovers patterns on its own
Example : Detect abnormal machine behavior automatically
AIoT Use Case : Anomaly detection , Device clustering , Behaviour analysis, Pattern discovery , Fraud detection



Reinforcement Learning :- Learning through rewards and penalties
Key Idea :- AI learns by interacting with environment & Improves continuously
Example : Smart cooling system optimizing energy usage
AIoT Use Case - Robotics, Autonomous vehicles, Smart traffic systems, Energy optimization , Industrial automation

Modern AIoT systems often combine all 3 techniques

Supervised Learning → Prediction
Unsupervised Learning → Anomaly Detection
Reinforcement Learning → Automation & Optimization